

MTH 732 Topology - Homework 8

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Assignment 1. Fix $n \geq 1$ and recall that $H_n(S^n) \simeq \mathbb{Z}$. Any continuous map $f : S^n \rightarrow S^n$ induces a map in n th homology $f_* : \mathbb{Z} \rightarrow \mathbb{Z}$ which is completely determined by $f_*(1)$. The integer $f_*(1)$ is called the degree of f , denoted $\deg f$.

- (a) Show that if f is a homeomorphism, then $\deg f = \pm 1$. In particular, the degree of the identity map is 1.
- (b) Show that if f is not surjective, then $\deg f = 0$.
- (c) The converse of (b) does not hold: a surjective map can have degree zero. Describe an example for $n = 1$.

Proof.

- (a) First consider when f is the identity on S^n . Then f_* is the identity on $\mathbb{Z}$, and so $f_*(1) = 1$ and the degree of f is 1.

Now if f is a homeomorphism, then let $a = \deg f$ and $b = \deg f^{-1}$. At the level of f , if we take a generator $[x] \in H_n(S^n)$, then we must have $f^{-1}(f([x])) = [x]$. Thus $(f^{-1})_*(f_*(1)) = 1$. We can view f_* as multiplication by the integer a , and we can view $(f^{-1})_*$ as multiplication by the integer b . Thus $(f^{-1})_*(f_*(1)) = ba = 1$, so b and a must be invertible. The only invertible elements in $\mathbb{Z}$ are 1 and -1, so these are our only options for $\deg f$ when f is a homeomorphism.

- (b) If f is not surjective, then there is at least one point $x \in S^2$ which is not in the image of f . Thus we can factor through $S^2 \setminus \{x\}$, which in homology corresponds to not being sent to a generator, and so $\deg f = 0$.
- (c) The idea here is that you want to send S^1 to a loop that goes around the circle once (or n times) and then back around the circle in the other direction once (or n times, respectively). This is surjective, but map is null-homotopic.

□

Assignment 2. Let $F = \{x_1, \dots, x_m\}$, $m \geq 1$, be a finite set of points on the sphere S^2 . Compute the relative homology groups $H_n(S^2, F)$ for all n .

Proof. We will heavily use the long exact sequence for the homology of a pair, along with the already-established fact (via simplicial homology) that $H_n(F) \cong \mathbb{Z}^m$ when $n = 0$ and $H_n(F) = 0$ otherwise. First, for $n \geq 2$, we use

$$H_n(F) \rightarrow H_n(S^2) \rightarrow H_n(S^2, F) \rightarrow H_{n-1}(F)$$

which becomes

$$0 \rightarrow H_n(S^2) \rightarrow H_n(S^2, F) \rightarrow 0,$$

showing that $H_n(S^2) \cong H_n(S^2, F)$. Thus

$$H_n(S^2, F) = \begin{cases} \mathbb{Z} & n = 2 \\ 0 & n > 2. \end{cases}$$

When $n = 1$, the same sequence with $H_{n-1}(S^2)$ added to the right end becomes

$$0 \rightarrow 0 \rightarrow H_1(S^2, F) \rightarrow \mathbb{Z}^m \rightarrow 0.$$

Thus $H_1(S^2, F) \cong \mathbb{Z}^m$.

Finally, when $n = 0$, take

$$H_0(F) \xrightarrow{i_*} H_0(S^2) \xrightarrow{\pi_*} H_0(S^2, F) \longrightarrow 0,$$

and consider the behavior of i_* . It takes any generator $[x] \in H_0(F)$ to the single generator of $H_0(S^2)$, since the path-connectedness of S^2 causes all the points of F to be identified in the quotient when viewed in S^2 . Thus i_* is surjective. Because the sequence is exact, this means $\ker \pi_* = H_0(S^2)$, so π_* is the zero map. From exactness on the final arrow, we also have that π_* is surjective, so we conclude that $H_0(S^2, F) = 0$.

We now have our final result:

$$H_n(S^2, F) = \begin{cases} \mathbb{Z}^m & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise.} \end{cases}$$

□

Assignment 3. Draw a small circle C on the surface of a torus T^2 in such a way that C separates T^2 into two connected components (so C bounds a disk). Compute the relative homology groups $H_n(T^2, C)$ for all n .

Proof. We will use the long exact sequence

$$H_n(C) \rightarrow H_n(T^2) \rightarrow H_n(T^2, C) \rightarrow H_{n-1}(C) \rightarrow H_{n-1}(T^2).$$

When $n \geq 3$, this becomes

$$0 \rightarrow 0 \rightarrow H_n(T^2, C) \rightarrow 0 \rightarrow 0,$$

so $H_n(T^2, C) = 0$.

For $n = 2$, we have

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\pi_*} H_2(T^2, C) \xrightarrow{f} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}^2.$$

Exactness gives us that π_* is injective, so we also know that $\text{im } \pi_*$ is a copy of $\mathbb{Z}$. Also, i_* sends the generator $[C]$ to zero since C bounds a disk, so i_* is the zero map and we get that f is surjective. This together with the fact that $\ker f = \text{im } \pi_* = \mathbb{Z}$ tells us that $H_2(T^2, C) = \mathbb{Z}^2$.

For $n = 1$, we have

$$\mathbb{Z} \xrightarrow{i_*} \mathbb{Z}^2 \xrightarrow{\pi_*} H_1(T^2, C) \xrightarrow{f} \mathbb{Z} \xrightarrow{j_*} \mathbb{Z}.$$

As above, we know that i_* is the zero map, so this tells us that π_* is injective. That also forces $\text{im } \pi_*$ to be a copy of $\mathbb{Z}^2$, which means $\ker f = \mathbb{Z}^2$. Since j_* is the induced map of inclusion for 0 homology, the generator gets sent to a generator, and thus j_* is an isomorphism. This means that $\text{im } f = \ker j_* = 0$. Thus $\mathbb{Z}^2 = \ker f = H_1(T^2, C)$.

Finally, for $n = 0$, we have

$$\mathbb{Z} \xrightarrow{i_*} \mathbb{Z} \xrightarrow{\pi_*} H_0(T^2, C) \xrightarrow{f} 0 \longrightarrow 0.$$

Since i_* is an isomorphism sending any generator to another generator, we have that $\ker \pi_* = \text{im } i_* = \mathbb{Z}$, and thus π_* is the zero map. Also, f must be the zero map, so $H_0(T^2, C) = \ker f = \text{im } \pi_*$. Putting those two facts together, we get $H_0(T^2, C) = 0$.

Thus

$$H_n(T^2, C) = \begin{cases} \mathbb{Z}^2 & n = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

□

Assignment 4. Redo the previous assignment, but this time take the circle C to be a non-trivial loop in T^2 (one of the two generators of $\pi_1(T^2)$). Your answer should be different. Generally speaking, this means that $H_n(X, A)$ can be different than $H_n(X, B)$ even if A and B are homeomorphic.

Proof. The only arguments this change affects are the arguments for $n = 1$ and $n = 2$, since those are where we encounter $i_* : H_1(C) \rightarrow H_1(T^2)$. Let a and b be the standard loops which serve as the generators of $H_1(T^2)$. Instead of being the zero map, since C does not bound a disk in T^2 , it gets sent to one of these generators, say a without loss of generality. Thus $i_* : \mathbb{Z} \rightarrow \mathbb{Z}^2$ is injective. We now proceed with the modified versions of $n = 1$ and $n = 2$ from above.

For $n = 2$, we have

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\pi_*} H_2(T^2, C) \xrightarrow{f} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}^2.$$

Exactness gives us that π_* is injective, so we also know that $\text{im } \pi_*$ is a copy of $\mathbb{Z}$, and thus $\ker f = \mathbb{Z}$. Since i_* is injective, we have $\text{im } f = \ker i_* = 0$, and thus $\ker f = H_2(T^2, C)$. Putting these two sentences together, we get $H_2(T^2, C) = \mathbb{Z}$.

For $n = 1$, we have

$$\mathbb{Z} \xrightarrow{i_*} \mathbb{Z}^2 \xrightarrow{\pi_*} H_1(T^2, C) \xrightarrow{f} \mathbb{Z} \xrightarrow{j_*} \mathbb{Z}.$$

Since i_* is injective, we have $\ker \pi_* = \text{im } i_* = \mathbb{Z}$. Thus $\ker f = \text{im } \pi_* = \mathbb{Z}$. We know that j_* is an isomorphism since it sends a generator of zero homology to a generator of zero homology, so $\text{im } f = \ker j_* = 0$. This leaves us to conclude that $H_1(T^2, C) = \mathbb{Z}$.

Thus

$$H_n(T^2, C) = \begin{cases} \mathbb{Z} & n = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

□

Assignment 5. Show that the spaces $X = S^1 \times S^1$ and $Y = S^1 \vee S^1 \vee S^2$ have the same homology groups in all dimensions, but are *not* homotopy equivalent.

Proof. Note that $X = T^2$. We know that these are not homotopy equivalent because $\pi_1(X) = \langle a, b : aba^{-1}b^{-1} = 1 \rangle$ and $\pi_1(Y) = \langle a, b \rangle$. The first is abelian and the second is not, so the fact that X and Y have different fundamental groups means that they are not homotopy equivalent.

We already know the homology of X , so we focus on Y . We will make use of the Mayer-Vietoris sequence

$$H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(Y) \rightarrow H_{n-1}(A \cap B)$$

where A is a slightly larger neighborhood of $S^1 \vee S^1$ within Y in the usual way, and B is a slightly larger neighborhood of S^2 , so that $A \cap B$ is null-homotopic, since we already know the homology of $S^1 \vee S^1$ from class. When $n \geq 2$, this becomes

$$0 \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(Y) \rightarrow 0,$$

so $H_n(Y)$ splits into $H_n(A) \oplus H_n(B)$.

When $n = 1$, we have

$$0 \rightarrow H_1(A) \oplus H_1(B) \xrightarrow{f} H_1(Y) \xrightarrow{g} \mathbb{Z} \xrightarrow{h} \mathbb{Z}^2,$$

where we have added an entry on the right for $H_0(A) \oplus H_0(B) = \mathbb{Z} \oplus \mathbb{Z} = \mathbb{Z}^2$. Exactness gives us that f is injective, which also tells us that $\ker g = \text{im } f = H_1(A) \oplus H_1(B)$. We also know that $h(1) = (1, -1)$, so h is injective, and we get $\text{im } g = \ker h = 0$. Thus $H_1(Y) = \ker g = H_1(A) \oplus H_1(B)$ and homology once again splits.

When $n = 0$, we can bypass all of this since Y is path-connected, and thus $H_0(Y) = \mathbb{Z}$. We can use this together with our knowledge of the homology of A and B write

$$H_n(Y) = \begin{cases} 0 \oplus 0 = 0 & n \geq 3 \\ 0 \oplus \mathbb{Z} = \mathbb{Z} & n = 2 \\ \mathbb{Z}^2 \oplus 0 = \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 0 \end{cases}$$

This does indeed agree with the homology of $X = T^2$, and we have already shown that X and Y are not homotopy equivalent. □